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Invictaeria: Aircraft Detection System

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Introduction

The Invictaeria project is a real-time aircraft detection and tracking system, developed as part of the Hardware course in the CTeSP in Cybersecurity at ISTEC Porto.

The system uses an ESP32 microcontroller connected to an OLED SSD1306 display (128×64) to show information about the nearest aircraft to the device's location. Data is obtained through the public OpenSky Network API, which provides real-time flight information based on ADS-B signals.

The name "Invictaeria" combines "Invicta" (a reference to the city of Porto) with "Aeria" (related to air), reflecting the nature and origin of the project.

Scope and Objectives

The main objective of the project was to build a portable device capable of:

- Connect to a Wi-Fi network and obtain the device's approximate location;
- Query the OpenSky Network API for aircraft data within an approximate 55 km radius;
- Display detailed information on the OLED screen about the nearest aircraft (ICAO24, callsign, latitude, longitude, altitude, speed, distance, and flight status);
- Visually indicate the aircraft's direction through a directional arrow calculated based on heading/bearing;
- Automatically update data every 30 seconds with a visible countdown timer.

The project was developed by a team of 4 students and presented as the final assignment for the Hardware course.

Components and Materials

The system was built with the following components:

Electronic Components

- ESP32 (with integrated Wi-Fi and Bluetooth);
- OLED SSD1306 128×64 display (I2C communication);
- Breadboard for prototyping;
- Jumpers (connection wires);
- Passive buzzer;
- LEDs (blue for Wi-Fi status, green for operation indication).

Enclosure Materials

- Cut and sanded wooden boards;
- Glass for the top lid (allowing interior viewing);
- TP-Link antennas (decorative/aesthetic);
- Hinges and metal latch;
- Black paint for finishing;
- Wood glue and double-sided tape;
- Wet sandpaper for splinter removal.

Architecture and Operation

The system's operational flow is as follows:

1. Initialization: The ESP32 initializes the OLED display and attempts to connect to the configured Wi-Fi network. An animated progress bar is displayed during connection.

2. Status indication: After successful Wi-Fi connection, the blue LED blinks 3 times and stays on. The name "Invictaeria" is displayed on the screen for 1.5 seconds.

3. Geolocation: The device obtains its approximate GPS coordinates through the ipinfo.io API (IP-based geolocation).

4. Aircraft query: Every 30 seconds, the system queries the OpenSky Network API requesting information on all aircraft within a ±0.5° latitude and longitude bounding box (~55 km).

5. Distance calculation: For each received aircraft, the distance to the device is calculated using the Haversine formula. The nearest aircraft is selected.

6. Display: The nearest aircraft's data is shown on the OLED screen: ICAO24, callsign, coordinates, altitude (meters), speed (km/h), distance (km), status (grounded/airborne), and a directional heading arrow.

7. Timer: A countdown timer in the upper right corner indicates how many seconds remain until the next update.

UML Diagram

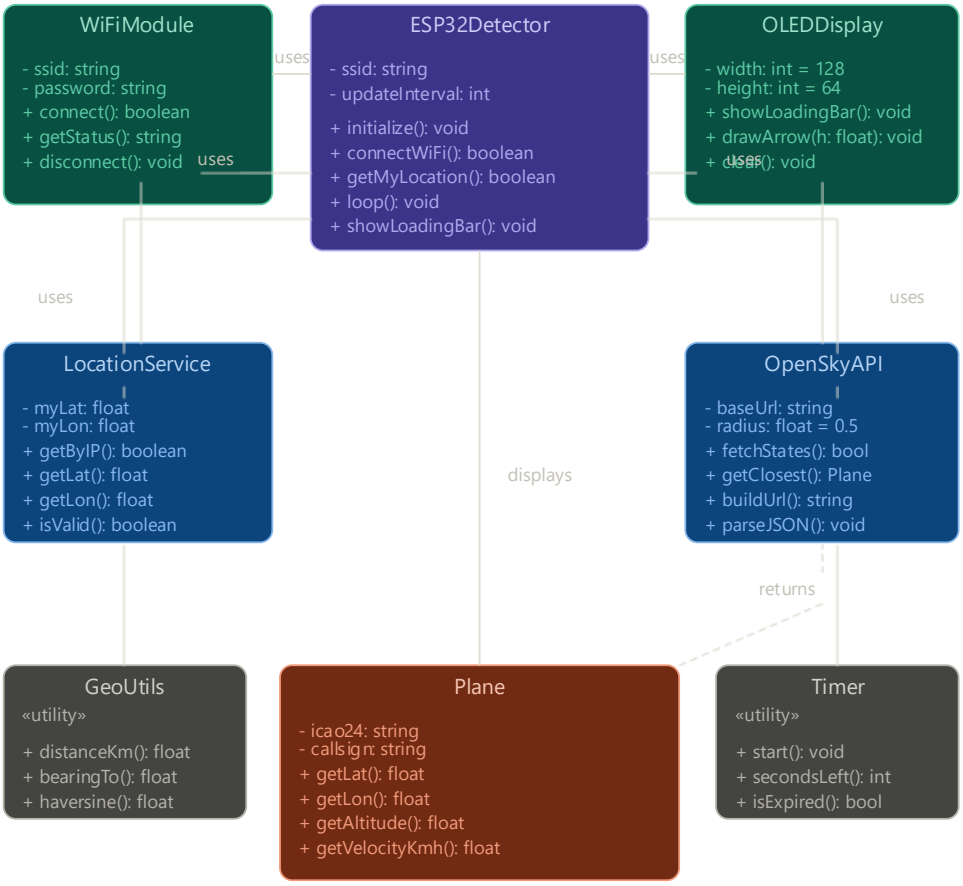

Figure 1: UML Class Diagram of the Invictaeria system1

Source Code

The code was developed in the Arduino IDE, using the following libraries:

- WiFi.h — Wi-Fi network connection;
- HTTPClient.h — HTTP requests to the API;
- ArduinoJson.h — JSON response processing;
- Wire.h — I2C communication with the OLED;
- Adafruit_GFX.h and Adafruit_SSD1306.h — OLED display control;
- math.h — Trigonometric functions for distance and bearing calculations.

The main functions implemented include:

- distanceKm() — Calculates the distance between two geographic points using the Haversine formula;
- drawHeadingArrow() — Draws a directional arrow on the OLED display based on the aircraft's heading angle;
- showLoadingBar() — Displays an animated loading bar during Wi-Fi connection;
- getMyLocation() — Obtains device coordinates via IP geolocation API;
- getPlaneData() — Queries the OpenSky Network API, selects the nearest aircraft, and displays data on the screen.

Repository Files

The project repository contains the following main files:

- final.ino — Final project code, used on the device. Implements IP geolocation (ipinfo.io), OpenSky Network API queries, Haversine distance calculation, directional arrow, and OLED display output;
- projetoESPFuncional.ino — Alternative version that uses a GPS Neo-6M module (via TinyGPS++) to obtain real coordinates instead of IP geolocation. Documents the project's evolution;
- conexão_oled_esp32.scss — ASCII pinout diagram describing the connections between the OLED SSD1306 display and the ESP32 (GND→GND, VCC→3.3V, SCL→GPIO22, SDA→GPIO21).

Physical Build

The device was assembled in a handcrafted wooden box built by the team. The box was cut, sanded, and painted black, with a glass lid that allows viewing the interior. TP-Link antennas were added as decorative elements, and a metal latch keeps the box closed. The following images document the construction process and the project's result.

Box Construction

Figure 2: Empty open wooden box2

Empty open wooden box, showing the handmade construction with hinges, glass lid, and black-painted interior finish.

Device Exterior View

Figure 3: Front view of the Invictaeria enclosure3

Front view of the Invictaeria enclosure with black wood finish, OLED display visible on top, decorative antennas, metal latch, and logo sticker.

Top View of Components with Enclosure Closed

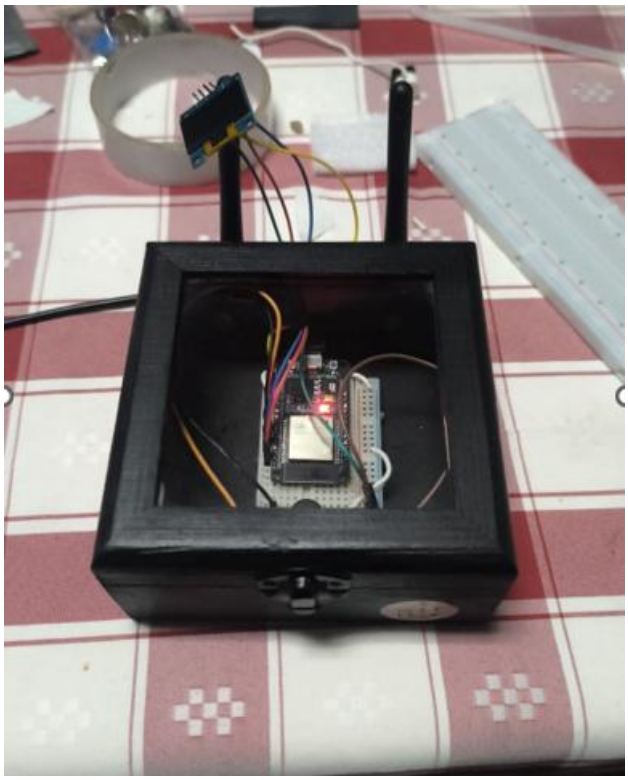

Figure 4: Top view of the open enclosure4

Top view of the open enclosure with all components assembled: ESP32, breadboard, jumper wires, OLED on top, and antennas.

Top View of Components with Enclosure Open

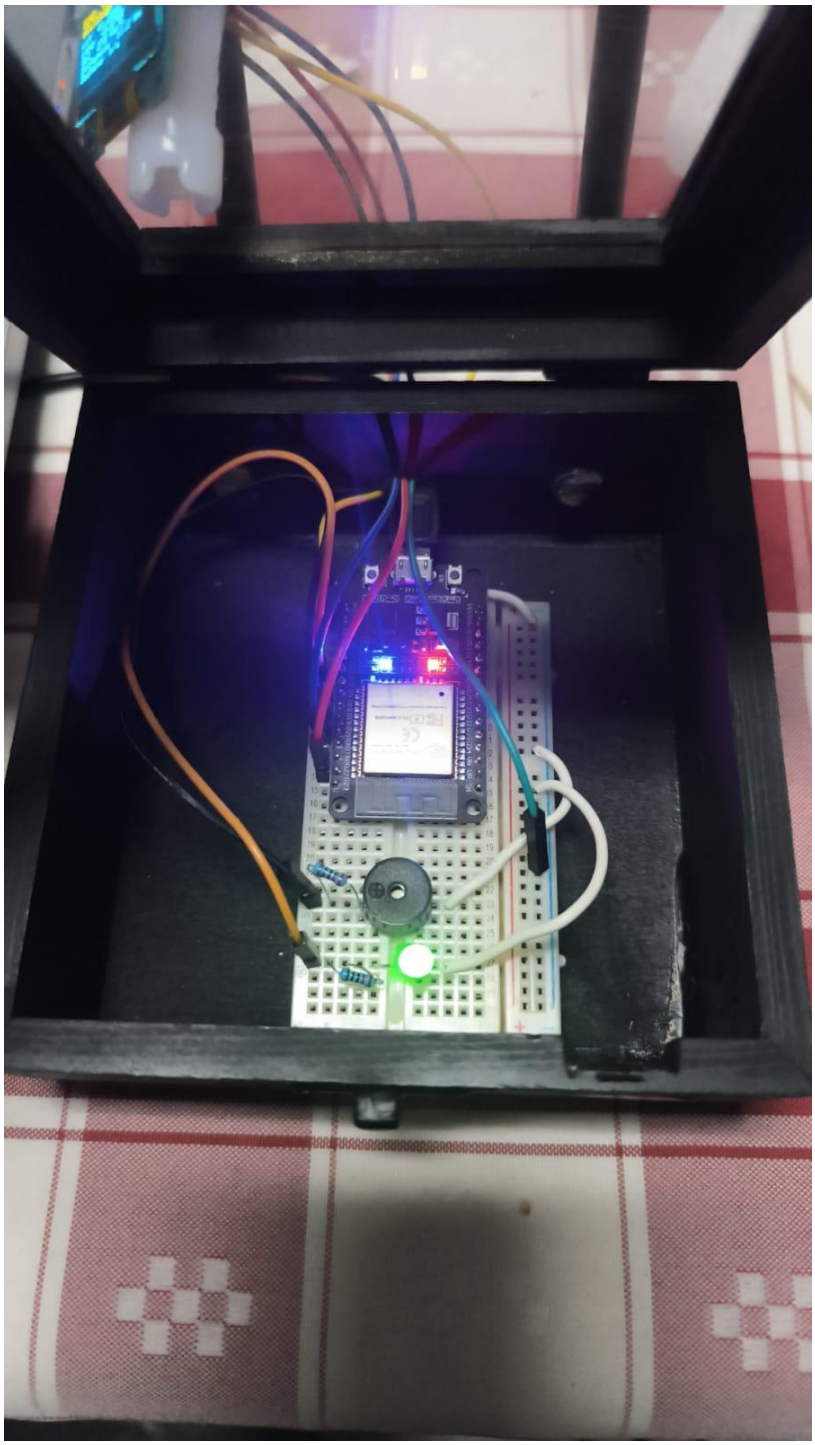

Figure 5: Device interior5

Device interior with the ESP32 mounted on a breadboard, passive buzzer, green LED lit, and jumper wires connecting to the OLED display.

System in operation

Figure 6: OLED display showing real-time flight data6

OLED SSD1306 display showing real-time flight data: ICAO 4d21ef, Callsign RYR96XD, coordinates, speed (278 km/h), distance (11.2 km), and directional arrow.

Challenges Encountered

Throughout the development of the project, the team encountered several challenges:

- GPS data interpretation: Understanding how the GPS module communicates with the ESP32 and correctly interpreting NMEA data proved more complex than initially anticipated.
- Directional arrow on OLED: Transforming the bearing angle into a coherent visual representation on the display required careful trigonometric calculations and multiple iterations.
- Display limitations: The 128×64 pixel OLED display is quite small, which limits the amount of information that can be shown at once.
- API rate limiting: The OpenSky Network API has usage limits that constrain the data update frequency.

Results

The tests performed demonstrated that the system works correctly and is capable of detecting aircraft in the surrounding area, presenting information legibly on the OLED screen.

The use of public APIs significantly simplified development and allowed obtaining reliable data without additional hardware (such as a dedicated ADS-B receiver).

The physical build — a black-painted wooden box with a glass lid, decorative antennas, and a metal latch — gave the device a professional and presentable appearance.

Future Improvements

The following possibilities for project evolution were identified:

- ADS-B receiver integration (via SDR) to capture signals directly from aircraft, eliminating API dependency;
- Data storage on SD card or transmission to an online database for flight history;
- FlightRadar24 API subscription for more detailed data (departure, arrival, airline);
- OLED display scroll implementation to show more information;
- Use of the GPS Neo-6M module for real location instead of IP geolocation.

Conclusion

The Invictaeria project demonstrated that it is possible to build a functional and compact aircraft detection system with accessible resources. The combination of the ESP32, OLED display, and public APIs made it possible to create a prototype that can be used both in an educational context and for personal curiosity.

The team acquired practical skills in embedded programming, I2C communication, REST API integration, geospatial calculations, and physical prototype construction.

Despite the identified limitations — namely API dependency and display constraints — the proposed objectives were successfully achieved.

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